

Building Block Design for Stacked Polyphase Bridge Converters for High-Voltage Traction Drives

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Abstract—Write last, 150–200 words, 4 numbers. (1) 800 V traction relies on 1.2 kV SiC; stacked polyphase bridges (SPB) split the bus into series low-voltage modules so that 100 V GaN HEMTs can be used. (2) The difficulty moves to the module: high current with paralleled devices, tight voltage margin. (3) We present a design methodology for a symmetric 75 V / 100 A_{rms} phase-leg building block with 2 × EPC2361 per switch and a centre gate driver, linking Q3D parasitic extraction to SPICE loss and gate-resistor design. (4) Results: 0.37 nH commutation loop, <0.1% left/right asymmetry, $R_{g,on}/R_{g,off} = 2/0 \Omega$ for ≤ 90 V stress, 99.58% estimated efficiency at 50 kHz; PCB copper is the second-largest loss.

Index Terms—Stacked polyphase bridge, GaN HEMT, paralleling, current sharing, PCB layout, parasitic extraction, DC-link capacitor, traction inverter.

I. INTRODUCTION

(0.75 page) *Problem, then idea, then why GaN, then the module challenge, then contributions.*

- 800 V traction trend: faster charging, lower current. State of the art: two-level 1.2 kV SiC. Limits: device FOM at high voltage, full-bus dv/dt on the winding insulation, one large module.
- SPB: the DC bus is split into N series-stacked modules; each module drives its own phase group of a multiphase machine and sees only V_{dc}/N .
- Why promising: 100 V GaN has a much better $R_{DS(on)} \cdot Q_G$ and no reverse recovery, so higher f_{sw} and efficiency; smaller voltage step per module; one repeated building block, modularity, fault tolerance, distributed cooling; multiphase machines.
- The challenge moves into the module: 100 A-class currents at 75 V need paralleled devices, and a 100 V device at 75 V leaves only about 15 V for overshoot. Layout, gate drive and current sharing decide whether the concept works.
- Contributions:
 - 1) Module requirements for an SPB derived from bus voltage, module count and device rating (75 V on 100 V GaN, 0.4 nH-class loop).
 - 2) A Q3D-to-SPICE design flow for paralleled GaN: copper-only partial-inductance extraction with component terminals, used directly for loss and gate-resistor design.
 - 3) A current-sharing sensitivity study with its mechanism (gate-loop mismatch acts as a delay comparable to the transition time; common-source in-

ductance imbalance is worst), which motivates the centre-driver symmetric layout.

- 4) Findings specific to low-voltage high-current modules: PCB copper as the second-largest loss, and DC-link sizing per floating module.

- Paper organisation sentence.

TODO: References: original SPB papers (Oxford, Rogers group); multiphase machines; 800 V SiC inverters; GaN traction inverters [1], [2], [3], [4], [5], [6].

II. STACKED POLYPHASE BRIDGE AND MODULE REQUIREMENTS

(0.5 page) *From system to module specification.*

- Fig. 1: SPB system: N stacked module DC links, each module drives a phase group of the machine.
- Module voltage: 75 V for 100 V devices, giving $N \approx 11$ for 800 V; voltage budget $75 + 15 \leq 90$ V, and $\Delta V = L_{loop} di/dt$ gives $L_{loop} \lesssim 0.4$ nH at about 20 A/ns.
- Current: 100 A_{rms} per phase (141 A peak); 2.65 kW per leg at $M = 1$; target 99.5% (13.3 W per leg).
- Table I: module specification.
- Note: legs per module (1 or 3) determines the DC-link ripple cancellation (Section V).

[have: reports/building-block/bb_report.tex (Sec. 1); scripts/bb_sizing.py] **TODO:** decide phases per module, N , machine data (phase count, winding sets).

III. BUILDING-BLOCK DESIGN

(1.25 page)

A. Device choice and paralleling

- EPC2361: 100 V, 0.75/1.0 mΩ, $Q_{rr} = 0$, 120 V transient rating; utilisation 75/100 vs 800/1200 for SiC.
- Two in parallel: conduction loss $I^2 R_{DS}/2 = 5.6$ W per leg at 100 °C, 35 A_{rms} per device; reference: EPC9186 uses the same 2 × EPC2361 for 130 A_{rms}.

B. Layout principles

- Vertical (inner-layer) commutation loop: L1 over L2 at 0.1 mm for flux cancellation.
- Two mirrored cells with the driver on the symmetry line; equal HO/LO stars; Kelvin returns; all terminals on the centre line.
- Stack-up: 4 layers, 2 oz; L2 DC−, L3 DC+, L4 AC. Fig. 2: top view and stack-up.

C. Parasitic extraction

- Method: copper only; MLCCs and FETs as gaps with source/sink terminals; partial R/L matrix exported as a SPICE subcircuit with coupling coefficients.
- Table II: commutation loop 0.312 nH (copper) / 0.374 nH (with ESL) per cell; gate loop 1.70 nH per FET; left/right difference $<0.1\%$.
- Sensitivity (one sentence or inset): prepreg 0.075/0.1/0.2 mm gives 0.30/0.34/0.50 nH.

[have: figures/layout_top.pdf, layout_stackup.pdf; simulation/bb_q3d; simulation/q3d (h_diel sweep)] *Ends with: the layout meets the 0.4 nH requirement of Section II with symmetric branches.*

IV. CURRENT SHARING AND SYMMETRIC GATE DRIVE

(1.25 page)

A. Mechanisms

Power-loop imbalance, gate-loop imbalance, common-source inductance (CSI), threshold spread [9], [7], [8], [1].

B. Sensitivity study

- Two EPC2361 in a double-pulse circuit, one parameter changed at a time (Fig. 3, three cases): gate loop 2→8 nH: 19 % dynamic imbalance, 4:1 E_{on} split; CSI imbalance (no Kelvin): 29 %; $\Delta V_{th} = 0.3$ V: 7 %, 2:1 E_{off} split.
- Mechanism: L_g/R_g delays the slower gate by about 2 ns, comparable to the 4 ns voltage transition, so one device takes most of the switching event; turn-off is load-driven and much less sensitive.
- Consequence: a side driver gives 2.0 vs 7.7 nH gate loops (Q3D) and is unacceptable; hence the centre driver with equal stars and Kelvin returns.

C. Result on the building block and gate-resistor selection

- LTspice with the Q3D subcircuits: 78.53 / 78.63 A at turn-off, overlapping waveforms (Fig. 4b).
- R_g selection at 160 A: $V_{DS} \leq 90$ V needs $R_{g,on} \geq 1.5 \Omega$, a gate bump ≤ 1 V needs 2Ω ; $R_{g,off} = 0$ (lowest E_{off} , least bump). Fig. 4a.

[have: simulation/sharing; simulation/dpt; simulation/bb_spice; figures rg_study, waveforms_141A] *Ends with: symmetric gate drive is a requirement, not a refinement.*

V. DC-LINK CAPACITOR DESIGN FOR STACKED MODULES

(0.75 page)

- Ripple current: single leg $\hat{I}/(2\sqrt{2}) = 50$ A worst, 28.8 A rated; three legs sharing a link (Kolar & Round [11]): 65 A worst, 50.3 A rated; checked with time-domain PWM.
- Capacitance: $\Delta V_{pp} \approx \hat{I}/(4f_{sw}C)$; Fig. 5: C vs ripple for 20–200 kHz; $\pm 2\%$ at 50 kHz: 78 μ F per leg if three legs share a module link, 236 μ F for a single-leg module.
- SPB point: each module has its own floating link, so only the legs inside one module cancel ripple; interleaving between stacked modules does not relieve the module capacitors.

- Selection: $24 \times 10 \mu$ F 1210 + $12 \times 1 \mu$ F 0805; per-capacitor current 0.9–2.1 A_{rms} from the Q3D-based current division.

[have: reports/dc-link; scripts/dclink_report.py; figures irms, creq]

VI. LOSSES AND EFFICIENCY

(0.5 page)

- Method: $E_{on}(i)$, $E_{off}(i)$ from LTspice with Q3D parasitics, averaged over the fundamental; conduction at 100 °C; copper from the Q3D DC matrix (load current) and a Q3D frequency sweep with a network solve at every switching harmonic.
- Fig. 6: breakdown vs f_{sw} . At 50 kHz: conduction 5.55, copper 3.30 (load current) + 1.07 (switching harmonics), switching 1.91, dead time 0.20, MLCC ESR 0.39, gate 0.08 W; total 12.5 W, 99.53 % (99.46 % with max $R_{DS(on)}$); the budget is reached at about 60 kHz.
- Insight: conduction + copper are about 80 % of the loss, and one fifth of the copper loss is at the switching harmonics (a point usually missed by DC-resistance estimates); with 1 oz inner layers copper alone would be 4.7 W, so DC-feed placement and copper weight are first-order design parameters in low-voltage high-current modules.

[have: figures/losses.pdf; reports/building-block/data/results.json]

VII. CONCLUSION AND FUTURE WORK

(0.5 page)

- Conclusion: mirror the four contributions with their numbers.
- Model credibility, without hardware: cross-check the extraction and models against known references: single-cell loop vs EPC's published optimal-layout range (0.35–0.4 nH at 4 mil); double-pulse model vs EPC2361 datasheet switching conditions; DC-link formulas vs Kolar & Round; component choice vs EPC9186. **TODO: decide which cross-checks to include (one short paragraph or a table).**
- Future work: double-pulse board at 75 V up to 160 A with per-device current shunts and Kelvin-referenced V_{GS} probing (overshoot vs R_g , E_{on}/E_{off} , dynamic sharing, high-side gate bump); routed-board re-extraction; two-module stacked demonstrator with DC-link voltage balancing; thermal validation.

FIGURE AND TABLE PLAN

Item	Content	Source
Fig. 1	SPB system: stacked modules + multiphase machine	TO
Fig. 2	Building-block top view + stack-up	layo
Fig. 3	Sharing sensitivity, 3 cases	Sha
Fig. 4	(a) R_g study (b) waveforms at 141 A	rg_s
Fig. 5	Ripple current vs M ; C vs ripple	irms
Fig. 6	Loss breakdown vs f_{sw}	loss
Table I	Module specification	bb_
Table II	Extracted parasitics	bb_
Table III	Component selection (optional)	bb_

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Full Rg grid; 1 oz/2 oz and DC-terminal placement study; capacitor placement variants (PL_A/B/C); local-bank/bulk resonance at 1.8/3.3 MHz; full 6-case sharing study; measured results.

REFERENCES

- [1] M. W. J. Cooke and D. J. Rogers, “120 V, 155 A traction inverter using eight parallel GaN HEMTs,” 2026. **TODO:** [venue](#)
- [2] “A high-performance GaN power module with parallel packaging for high-current and low-voltage traction inverter applications.” **TODO:** [authors/venue](#)
- [3] “Design considerations of a GaN-based three-level traction inverter for electric vehicles.” **TODO:** [authors/venue](#)
- [4] “Systematic efficiency-density co-optimization of 100 kW GaN traction inverter: methodology and integration.” **TODO:** [authors/venue](#)
- [5] “GaN-based low-voltage inverter for electric scooter drive system.” **TODO:** [authors/venue](#)
- [6] “A multi-kilowatt low-profile GaN inverter for light electric vehicles and high-power tools.” **TODO:** [authors/venue](#)
- [7] “Parallel connection of GaN FETs: an experimental investigation approach.” **TODO:** [authors/venue](#)
- [8] “Paralleling GaN E-HEMTs in 10 kW–100 kW systems.” **TODO:** [authors/venue](#)
- [9] D. Reusch, “Effectively paralleling gallium nitride transistors for high current and high frequency applications,” EPC AN020.
- [10] EPC, “EPC2361 datasheet,” rev. 2.4, 2026.
- [11] J. W. Kolar and S. D. Round, “Analytical calculation of the RMS current stress on the DC-link capacitor of voltage-PWM converter systems,” *IEE Proc. Electr. Power Appl.*, 2006.